substance for their activities and incorporate into their protoplasm, this process is called assimilation.

5.4 Egestion

The digestive waste and unabsorbed substances in the ileum enter into the large intestine and it mostly contains fibre called roughage. The roughage is utilized by symbiotic bacteria in the large intestine for the production of substances like vitamin K and other metabolites. All these substances are absorbed in the colon along with water. The waste is then solidified into faecal matter in the rectum. The faecal matter initiates a neural reflex causing an urge or desire for its removal. The egestion of faeces through the anal opening is called defaecation. It is a voluntary process and is carried out by a peristaltic movement.

5.5 Nutrients, Vitamins and Minerals

Food comprises of macronutrients and micronutrients. The nutrients required in larger quantities are called macronutrients, whereas those required in small quantities are called micronutrients. Essential nutrients cannot be synthesized by the body; they have to be included in the diet. Macronutrients are lipids, carbohydrates, proteins and the micronutrients are vitamins and minerals. Water plays an important role in the metabolic processes and prevents dehydration of the body.

Intake of too much of food or lesser amount of food than the basic requirement is called **malnutrition**. A diet which can provide all the metabolic requirements of the body in a right proportion is called **balanced diet**. That means it should contain carbohydrates and fats for energy yielding, proteins for growth and replacement; and vitamins, minerals and water for physiological regulation.

Vitamins

Vitamins are naturally occurring organic substances regularly needed in minute quantities for maintaining normal health as metabolic regulators. The identified vitamins are classified as fat soluble (A,D,E and K) (Table 5.1) and vitamin B and vitamin C are water soluble (Table 5.2). Vitamin A, D, E and K, if consumed beyond required level may cause defects, commonly referred to as hypervitaminosis.



N.I. Lunin discovered vitamins but the name vitamin was given by Dr. Funk (1912). The first vitamin isolated was B1 by Dr.Funk. The first vitamin produced by fermentation process using, *Acetobacter* bacteria is Vitamin C.

Minerals

These are the inorganic chemical elements, i.e., Ca, Fe, I, K, Mg, Na, P, S, etc needed for regulation of various physiological functions. These can be classified into major minerals (Na, P, K, Ca, Mg, S, Cl) and others are trace minerals such as Fe, Cu, Zn, Co, Mn, I, and fluorine. Sodium ions are more abundant than any other cation in the body fluids.

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